

Exploring the Impacts of Interactive Tasks on Redirected Walking in Virtual Reality

Jieun Lee *, Aya Ataya †, SeungJun Kim ‡

School of Integrated Technology, Gwangju Institute of Science and Technology (GIST)

ABSTRACT

Reorientation in virtual reality (VR) is manipulating user orientation within a small physical space to support travel in a large virtual space. Reorientation can be applied to alter the user's orientation in VR content as the user rotates to engage with the content. This study integrated different types of interactions with reorientation to reduce the discrepancy between virtual and real rotation without causing user discomfort. We exceeded the rotation gain threshold to examine the enhancement of reorientation performance in 48 participants. The results indicate that engagement in interactive tasks effectively prevents users from discerning reorientation manipulation without inducing VR sickness.

Keywords: Redirected walking, Reorientation, Interactivity, User experience.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

The reorientation technique is a software-based method that alters the orientation of a user wearing a virtual reality (VR) headset [1]. Its primary function is to ensure the safety of VR users by preventing them from leaving the tracking area or colliding with objects. This is achieved by manipulating the user's orientation within a virtual environment at a specific rate called the rotation gain, which alters the perceived extent of the user's rotation [2]. Although a large rotation gain can improve reorientation performance, an excessive visual-vestibular mismatch may cause discomfort or VR sickness [3]. Accordingly, the rotation gain is recommended to be maintained within a detection threshold (DT), which is estimated experimentally [2].

Recent studies have attempted to improve the DT to increase reorientation performance without causing user discomfort, with strategies including the use of distractors [4] and the involvement of users in interactions [5]. Although these attempts provide valuable insights into interactivity within VR environments, further research is required to improve user experience in applications employing the reorientation technique with a particular focus on the diversity of interactions. The present study's objective was to evaluate reorientation performance and user comfort stemming from the integration of different VR interactions using a reorientation technique. To effectively

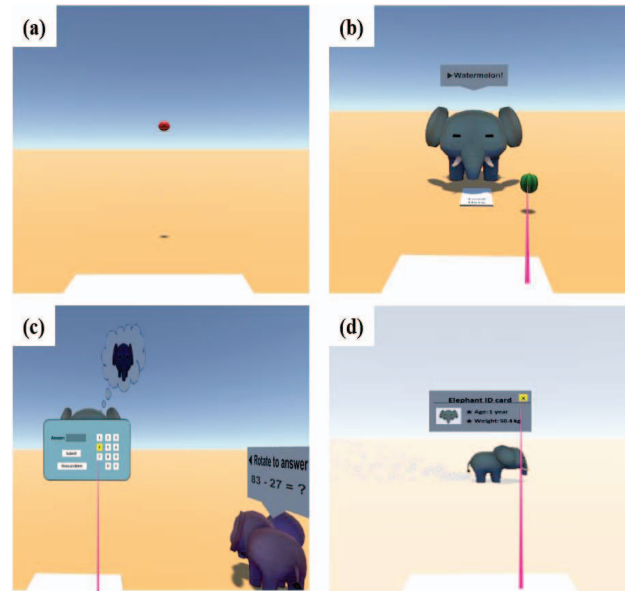


Figure 1: A sequence of snapshots from a participant's viewpoint during interactive tasks. Following a 90-degree rotation, the participant is (a) observing a red ball, (b) picking up a watermelon to give the elephant, (c) solving an arithmetic problem, and (d) checking a message. Concurrently, the participant rotates back to their initial orientation in the virtual environment.

evaluate the improvement in reorientation performance, we reoriented users with rotation gains exceeding the DT.

2 USER STUDY

Interactive task design: In this experiment, we designed four different interactive tasks using Unity. The virtual context in which the participants performed these tasks was that of a zookeeper caring for an elephant. Figure 1(a) depicts a non-interactive (NoInt.) scenario wherein the participants observed a red ball that appeared in the virtual environment. In the pickup and place (P/P) task, participants used a VR controller to pick up a watermelon and feed it to the elephant (see Fig. 1(b)). As shown in Figure 1(c), the 'Quiz' task had participants solving math problems presented by the elephant and submitting their answers. In the 'Message' task, participants read and closed a message window carried by the elephant (see Figure 1(d)). The elephant's movement created dust, which obstructed the participants' vision.

Rotation gain: Rotation gain is the ratio between virtual and real rotation. For example, if a rotation of 90° in the VR environment is 180° in reality, the rotation gain is 0.5. For this experiment, we used a rotation gain of 0.7, close to the DT of 0.67 [2], as well as two greater rotation gains of 0.6 and 0.5, to examine the improvement in reorientation performance.

* jieunlee@gm.gist.ac.kr

† aya.ataya@gm.gist.ac.kr

‡ seungjun@gist.ac.kr

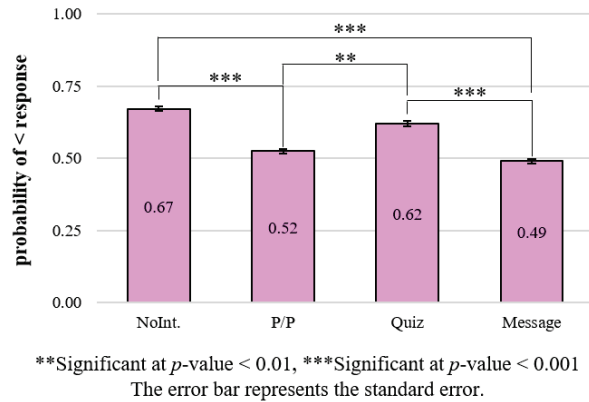


Figure 2: Participant perception analysis in the discrimination between virtual and physical rotation: Probability of 'smaller' responses by interactive task type.

Procedure: Participants initially walked for 4 m while wearing the VR headset, and were asked to rotate 90° to the right or left. To complete the interactive task, the participants had to rotate by 90° in the opposite direction. The rotation gain was applied after the first 90-degree turn and deactivated after the second turn. Upon completion of this task, the participants answered a two-alternative forced choice (2AFC) questionnaire ('Was the virtual rotation smaller or greater than the physical rotation?') [2]. Each task was repeated three times, after that followed by a VR motion sickness questionnaire [6]. We adopted a balanced-Latin-square-based study design and recruited 48 participants (age ($M = 21.83$, $SD = 2.10$)). The experiment was conducted over two hours, and each participant was rewarded with \$30 for their time and effort.

3 RESULTS

We performed a repeated analysis of variance across the four interactive tasks to estimate the probability that the participants would respond 'smaller' to the 2AFC question and examine the VR sickness score. Post-hoc tests were performed using Bonferroni adjustments for each measurement.

Figure 2 shows a significant relationship between interactive task type and probability of 'smaller' responses, with $F(3, 423) = 14.23$, $p < 0.001$. Significant differences were observed between the Message condition and NoInt. and Quiz conditions ($p < 0.001$). In addition, the P/P condition was associated with a lower response probability than that of the NoInt. ($p < 0.001$) and Quiz ($p = 0.006$) conditions. Participants tended to be less aware of reorientation manipulation in the P/P and Message conditions than in the NoInt. condition. A similar trend was observed for the Quiz condition, although the difference was not statistically significant.

The VR sickness scores reported by the participants across all interactive tasks were similar and relatively low compared to 100 points (see Figure 3). Our scenarios did not appear to contribute significantly to VR sickness. This finding suggests that integrating reorientation techniques into interactive tasks does not raise concerns regarding the induction of VR sickness.

4 DISCUSSION AND CONCLUSION

This study was conducted to investigate whether integrating the reorientation technique into interactive VR tasks improves reorientation performance and user experience. The results indicate that participants did not easily notice the effect of reorientation manipulation, even when the rotation gain was applied beyond the established DT. Furthermore, we found that when the reorientation technique was integrated with interactive

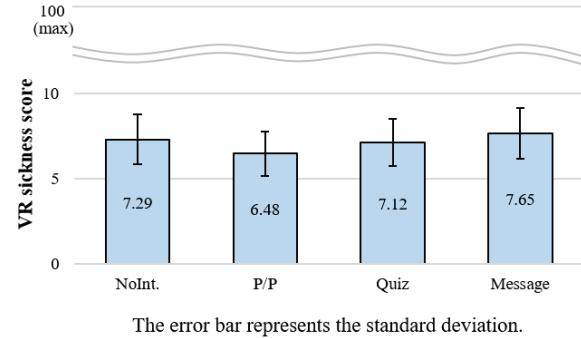


Figure 3: VR sickness scores according to interactive task type.

tasks, the level of VR sickness experienced by the participants did not increase under any condition. These results suggest that rotation gains exceeding the DT might not induce user discomfort if integrated with interactive tasks. In future studies, this trend can be examined through more diverse applications, as well as analyses of physiological data and qualitative evaluation results. Ultimately, real-time monitoring can assist in increasing the reorientation effect without disturbing the user's engagement in VR content.

ACKNOWLEDGMENTS

This work was partly supported by the GIST-MIT Research Collaboration grant funded by the GIST in 2023 and the Culture, Sports and Tourism Research and Development Program through the Korea Creative Content Agency funded by the Ministry of Culture, Sports and Tourism, in 2022, through the Project Name: Development of Artificial Intelligence-Based Game Simulation Technology to Support Online Game Content Production under Project R2022020070.

REFERENCES

- [1] Bruder, G., Steinicke, F., Hinrichs, K. H., & Lappe, M. (2009, December). Reorientation During Body Turns. In EGVE/ICAT/EuroVR (pp. 145-152).
- [2] Steinicke, F., Bruder, G., Jerald, J., Frenz, H., & Lappe, M. (2009). Estimation of detection thresholds for redirected walking techniques. IEEE transactions on visualization and computer graphics, 16(1), 17-27.
- [3] Akiduki, H., Nishiike, S., Watanabe, H., Matsuoka, K., Kubo, T., & Takeda, N. (2003). Visual-vestibular conflict induced by virtual reality in humans. Neuroscience Letters 340(3), 197-200.
- [4] Peck, T. C., Fuchs, H., Whitton, M. C. (2009). Evaluation of reorientation techniques and distractors for walking in large virtual environments. IEEE transactions on visualization and computer graphics, 15(3), 383-394.
- [5] Schmelter, T., Hernadi, L., Störmer, M. A., Steinicke, F., Hildebrand, K. (2021). Interaction-based redirected walking. ACM Proceedings on Computer Graphics and Interactive Techniques, 4(1), 1-16.
- [6] Kim, H. K., Park, J., Choi, Y., & Choe, M. (2018). Virtual reality sickness questionnaire (VRSQ): Motion sickness measurement index in a virtual reality environment. Applied Ergonomics, 69, 66-73.